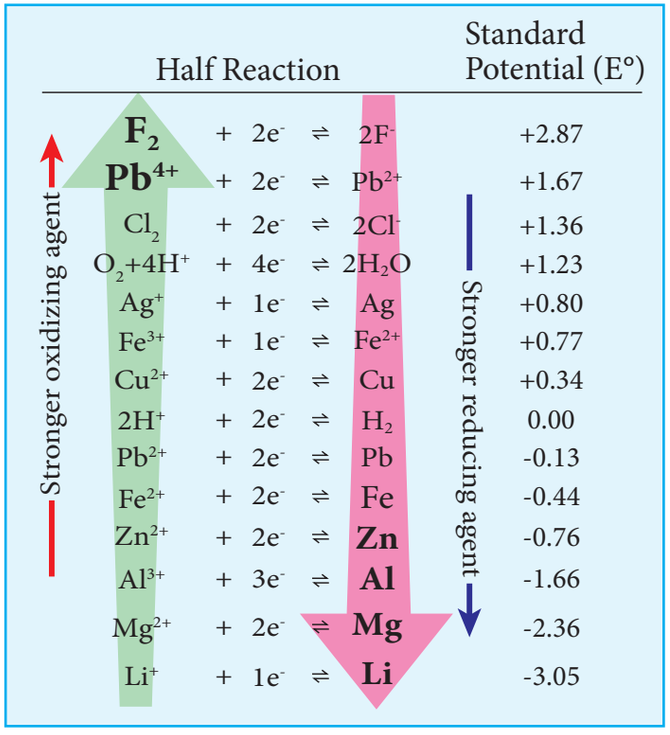
## Electrochemical series

We have already learnt that the standard single electrode potentials are measured using standard hydrogen electrode. The standard electrode potential at 298K for various metal - metal ion electrodes are arranged in the decreasing order of their standard reduction potential values as shown in the figure.

This series is called electrochemical series.

The standard reduction potential ($E^\circ$) is a measure of the oxidising tendency of the species. The greater the $E^\circ$ value, greater is the tendency shown by the species to accept electrons and undergo reduction. So higher the ($E^\circ$) Value, lesser is the tendency to undergo corrosion




## EVALUATION

### Choose the correct answer:

1. The number of electrons that have a total charge of 9650 coulombs is

   a) $6.22\times10^{23}$ b) $6.022\times10^{24}$

   c) $6.022\times10^{22}$ c) $6.022\times10^{-34}$

2. Consider the following half cell reactions:

   $Mn^{2+} + 2e^- \rightarrow Mn \quad E^\circ = -1.18V$

   $Mn^{2+} \rightarrow Mn^{3+} + e^- \quad E^\circ = -1.51V$

   The $E^\circ$ for the reaction $3Mn^{2+} \rightarrow Mn + 2Mn^{3+}$, and the possibility of the forward reaction are respectively.

   a) 2.69V and spontaneous b) -2.69 and non spontaneous

   c) 0.33V and Spontaneous d) 4.18V and non spontaneous

3. The button cell used in watches function as follows

   $Zn\,(s) + Ag_2O\,(s) + H_2O\,(l) \rightleftharpoons 2\,Ag\,(s) + Zn^{2+}(aq) + 2OH^-\,(aq)$ the half cell potentials are $Ag_2O\,(s) + H_2O\,(l) + 2e^- \rightarrow 2Ag\,(s) + 2\,OH^-\,(aq) \quad E^\circ = 0.34V$ and $Zn\,(s) \rightarrow Zn^{2+}(aq) + 2e^- \quad E^0 = 0.76\,V$. The cell potential will be

   a) 0.84V b) 1.34V c) 1.10V d) 0.42V

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